

GENE REGULATORY NETWORK INFERENCE FROM INTEGRATIVE MULTI-OMICS DATA

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Problem Statement

Accurately mapping GRNs from simultaneous single-cell multi-omics data, specifically joint scRNA-seq and scATAC-seq

Paper 1 - ATFGRN

Uses only scRNA-seq data

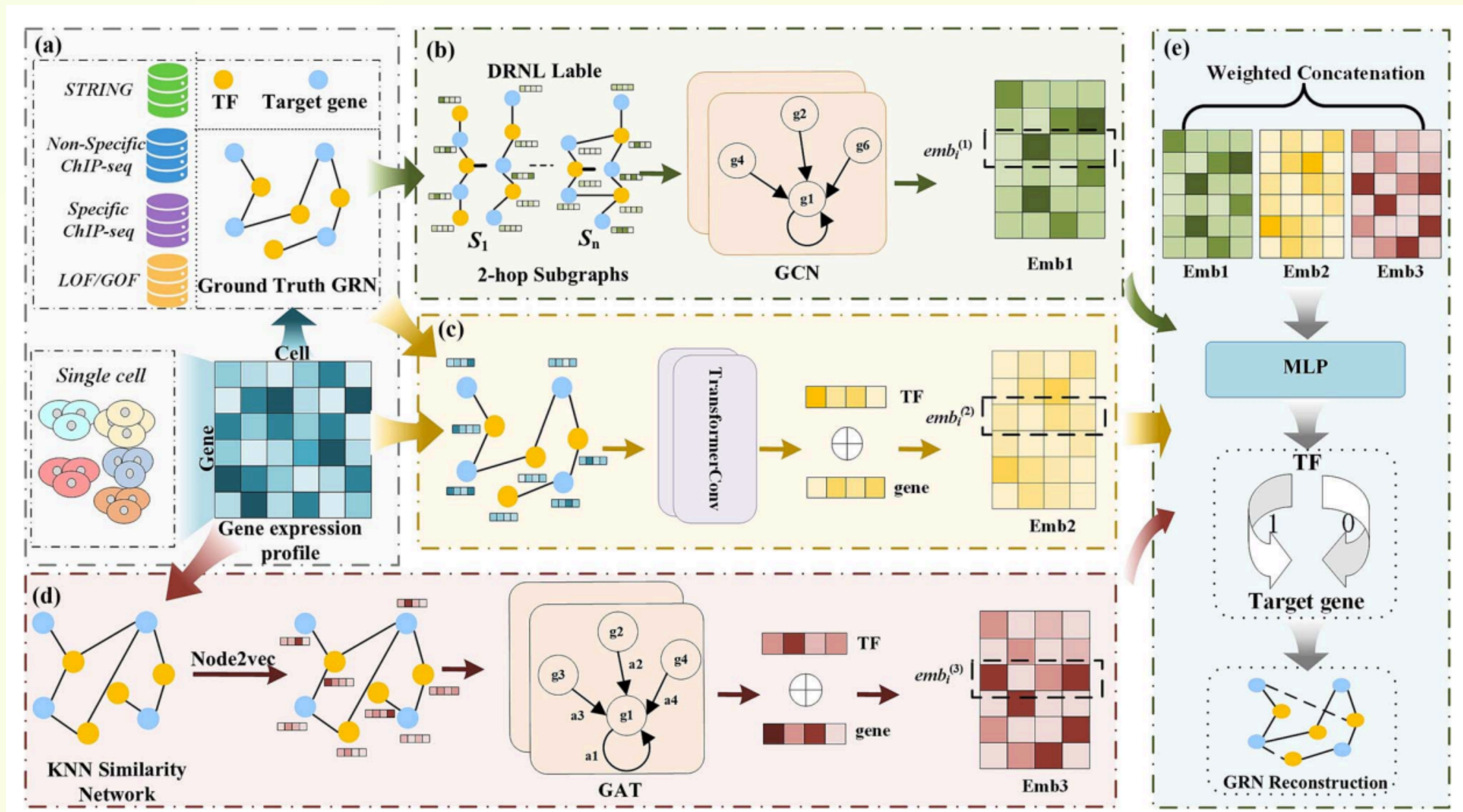
Datasets & Ground Truth

- **Primary Datasets:** Single-cell transcriptomic datasets (scRNA-seq), including well-established benchmark datasets like hESC (human embryonic stem cells) and mDC (mouse dendritic cells), evaluated at varying scales (e.g., 500 to 1,000 cells).
- **Input Ground Truth (Training):** Gene-cell expression matrices paired with initial structural regulatory networks (prior GRNs) to initialize the graph topologies.
- **Validation Ground Truth (Testing):** Experimentally validated cell-type-specific and non-specific networks (often aligned with frameworks like BEELINE, utilizing ChIP-seq or knockout data) to benchmark true regulatory edges.

Training & Testing Methodology

- **Training Process:** The model is optimized using a PyTorch/PyTorch Geometric backend, operating over independent runs with standard deep learning hyperparameter tuning (Adam optimizer, batch size of 32, ~400 epochs, and learning rate of 0.001).
- **Testing Phase:** Evaluates the model's ability to predict unseen regulatory links. The testing methodology rigorously assesses cross-network generalization and the model's structural resilience to simulated zero-inflation (dropout noise) prevalent in scRNA-seq matrices.

Model Architecture



Benchmarks & Results

- **Baseline Models:** Extensively benchmarked against mainstream GRN inference algorithms, including traditional methods (GENIE3, GRNBoost2, SCENIC) and deep learning/GNN approaches (CNNC, DGRNs, GENELink, scMGATGRN).
- **Evaluation Metrics:** Systematically measured using AUROC (Area Under the Receiver Operating Characteristic) and AUPRC (Area Under the Precision-Recall Curve).
- **Key Results:** ATFGRN demonstrated superior predictive capability, improving AUROC performance by 5.09% over existing state-of-the-art approaches across four distinct types of regulatory networks, successfully mitigating GNN over-smoothing.

Paper 2 - *scMultiomeGRN*

Integration of scRNA-seq and scATAC-seq data

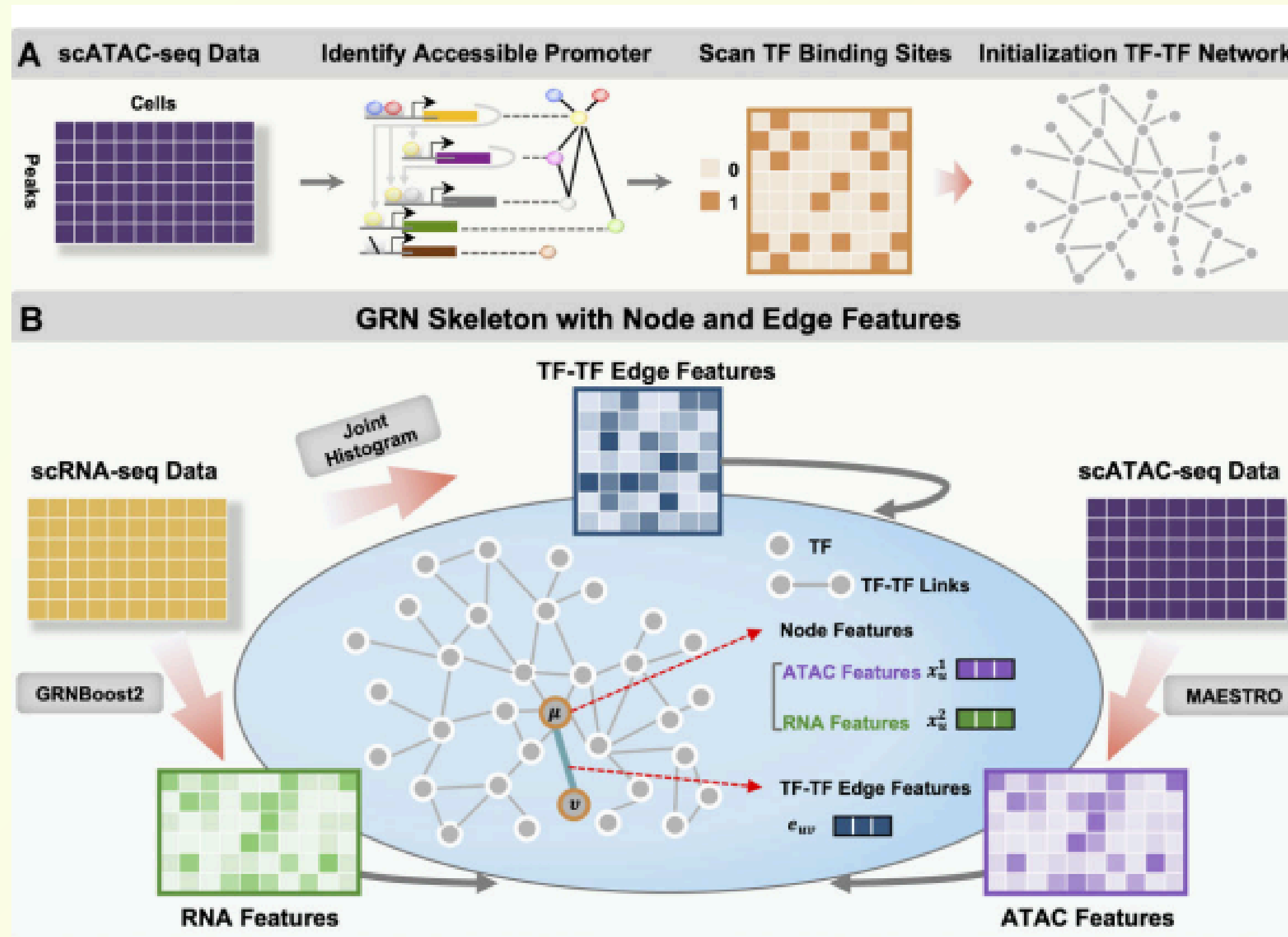
Datasets & Ground Truth

- **Primary Datasets:** Evaluated on diverse single-cell multiome datasets, including Peripheral Blood Mononuclear Cells (PBMCs) containing ~7,189 ATAC and ~9,129 RNA cells (spanning subsets like CD14+ Monocytes), and a massive Alzheimer's Disease (AD) atlas covering 414,000 snRNA-seq and 437,000 snATAC-seq nuclei across varying disease stages.
- **Input Ground Truth (Prior Graph):** Utilizes chromatin accessibility data (scATAC-seq) to establish an initial adjacency matrix (prior graph), acting as the foundational map of potential physical binding interactions between Transcription Factors (TFs) and target genes.
- **Validation Ground Truth:** Experimentally validated biological databases (such as non-specific ChIP-seq networks or STRING database functional interactions) to rigorously benchmark the structural accuracy of the predicted edges.

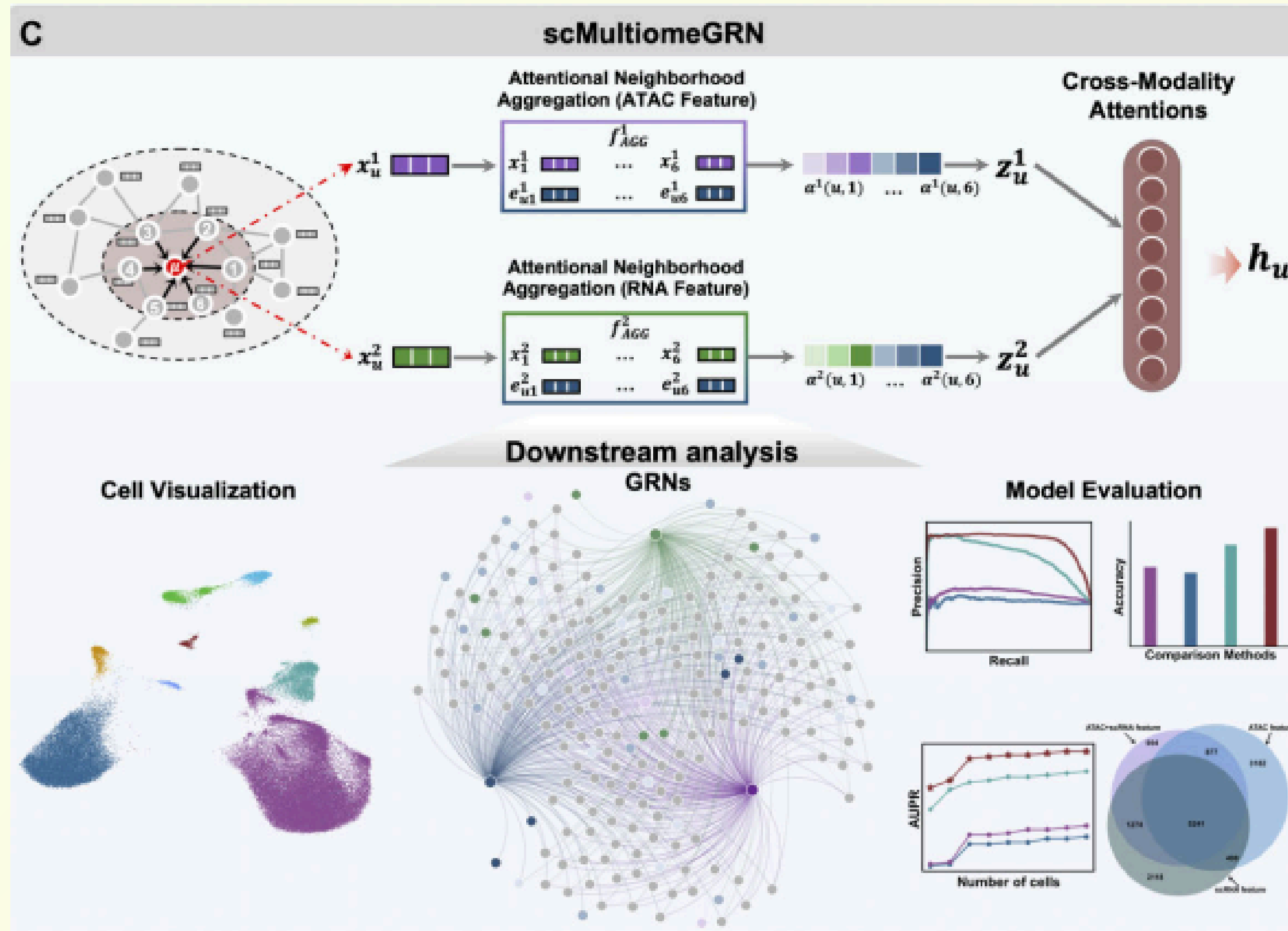
Training & Testing Methodology

- **Data Preprocessing:** Operates by first independently processing scRNA-seq and scATAC-seq matrices, managing dropouts, and aligning the modalities to establish a unified single-cell view.
- **Model Optimization:** Trained using a deep learning pipeline designed to learn cross-omics non-linear correlations through advanced graph learning and attention mechanisms, mapping latent states to observed network topologies.
- **Inference Scale:** Evaluated not merely at a generalized population level, but tested extensively on extracting complex regulatory mechanisms for distinct cell sub-types and disease states (e.g., early vs. late-stage Alzheimer's Disease), where the latter was done as a case study.

Model Architecture



Model Architecture [Contd]



Benchmarks & Results

- **Baseline Models:** Extensively benchmarked against state-of-the-art computational GRN models, including structural equation models with beta-VAE frameworks (DeepSEM) and co-expression convolutional neural networks (CNNC).
- **Evaluation Metrics:** Standard structural prediction metrics assessing the model's ability to accurately recover complex regulatory interactions, particularly on rare cell types or subtypes with limited cell counts.
- **Key Results:** scMultiomeGRN significantly outperformed competing state-of-the-art models on multiple benchmark datasets involved in disease and health, proving highly resilient to the high loss rates characteristic of single-cell sequencing.

Paper 3 - scTFBridge

Integration of scRNA-seq and scATAC-seq data

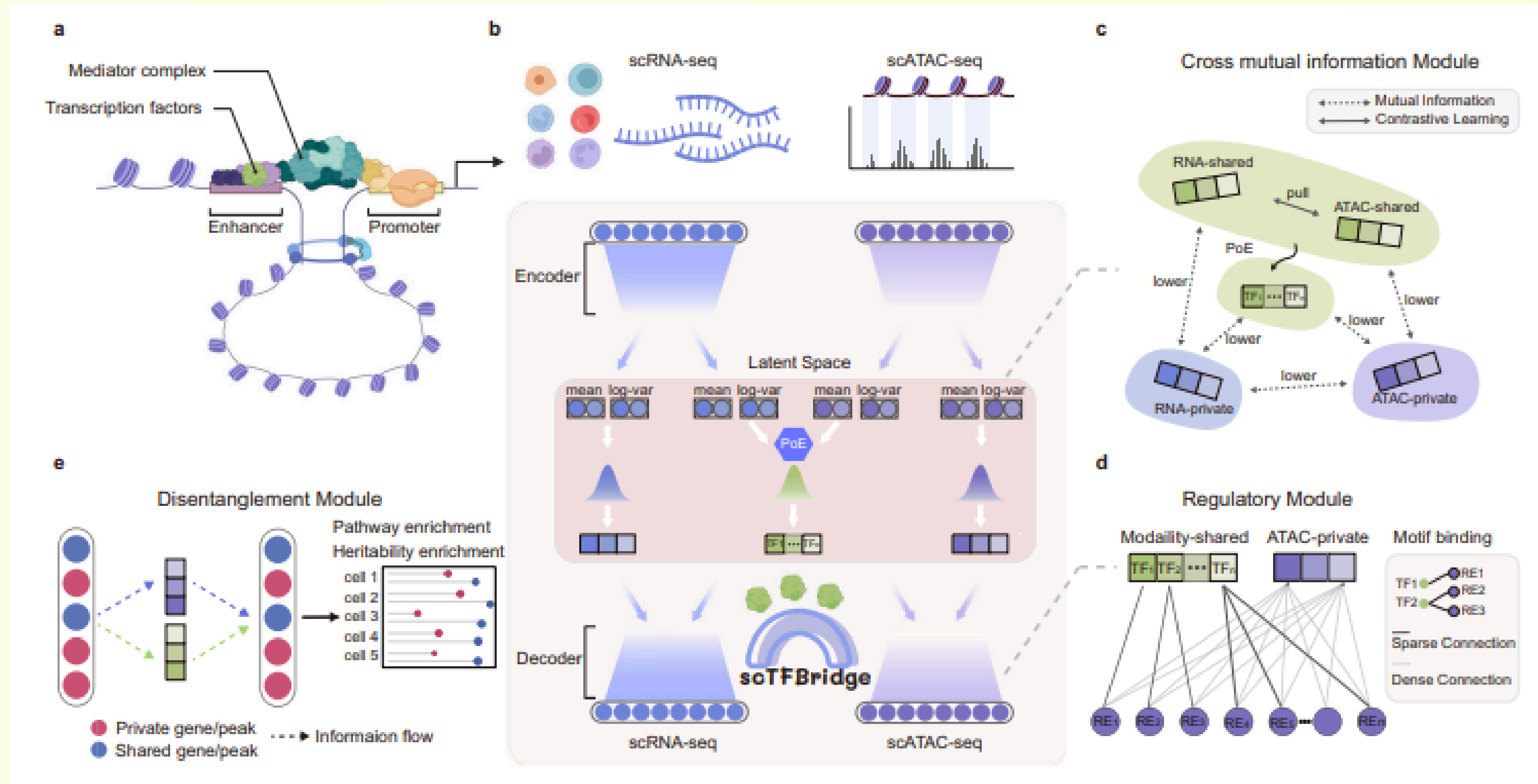
Datasets & Ground Truth

- **Primary Datasets:** Sourced from 10x Genomics Multiome platforms, including Bone Marrow (BMMC) with 171 to 2,168 cells per type , Peripheral Blood (PBMC) with 25 to 511 cells per type , and Lung Cancer biopsies from 16 distinct donors.
- **Input Prior (For Training):** Sequence-derived TF-motif binding matrices retrieved from JASPAR and CIS-BP databases, utilizing 713 TF Position Weight Matrices (PWMs) to structurally constrain the network.
- **Validation Ground Truth (For Testing):** Experimentally verified physical bindings from ChIP-seq consortiums (Cistrome, ENCODE, ReMap) and multi-omic CRISPR perturbation screens.

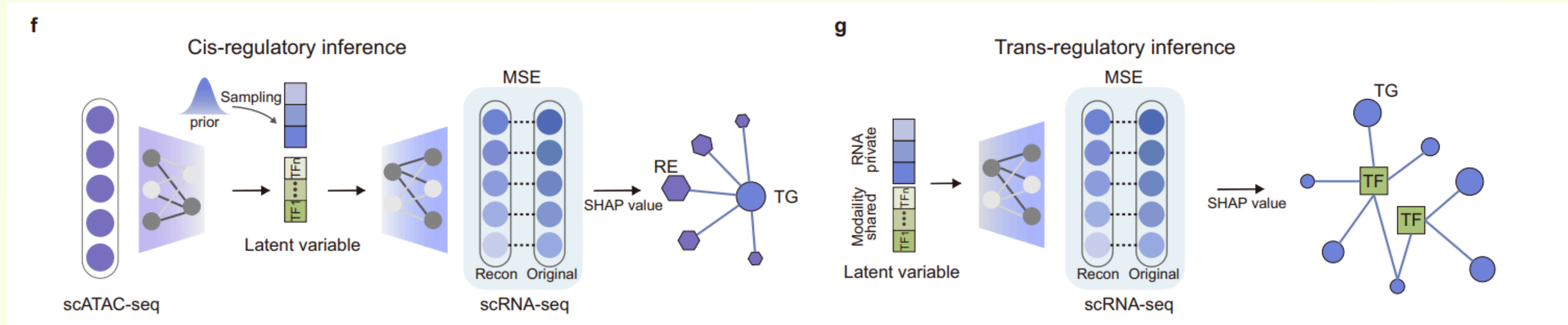
Training & Testing Methodology

- **Data Split:** The dataset utilized a 5-fold cross-validation approach with a strict separation of 64% for training, 16% for validation, and 20% for testing.
- **Training Process:** The model was optimized using the Adam optimizer with a batch size of 128 across 150 epochs. The 16% validation set was used for early stopping and hyperparameter tuning.
- **Testing Phase:** The actual inference (extracting the final GRN via SHAP value computation) was strictly isolated to the 20% test datasets to guarantee unbiased evaluation.

Model Architecture



Inference Methodology (Graph Generation)



- **Simulation Engine:** Post-training, the model uses deep learning explainability techniques, specifically SHAP, to track feature attribution.
- **Gradient Calculation:** It calculates the mathematical gradient (derivative) of a predicted Target Gene's expression with respect to the input Regulatory Elements and latent TF nodes.
- **Network Extraction:** By nudging a TF and observing the ripple effect on 3,000 highly expressed target genes, the model extracts discrete regulatory scores to draw the final TF-> RE->TG graph.

Benchmarks & Results

- **Baseline Models:** Evaluated against state-of-the-art multi-omic tools including LINGER, SCENIC+, scMTNI, and scMultiomeGRN.
- **Evaluation Metrics:** Measured via AUROC (Area Under the Receiver Operating Characteristic) and AUPRC (Area Under the Precision-Recall Curve).
- **Key Results:** scTFBridge demonstrated highly robust multi-omic batch effect isolation and significantly outperformed all baselines in predicting both cis- and trans-regulatory elements. Improvements were statistically significant, with p-values frequently < 0.01 and as low as 10^{-6} .

Note: *Traditional scRNA-seq benchmarks like BEELINE were incompatible due to the dual-input requirement*

Next Steps

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SC-MO-GRN-DB

- **Comprehensive Repository:** A centralized database tracking single-cell multi-omic datasets specifically for single-cell gene regulatory networks (GRNs).
- **Experimental Baselines:** Uses high-confidence physical evidence (ChIP-seq localization) and functional evidence (CRISPR/shRNA perturbations) rather than computational predictions.
- **Strict Cell-Type Matching:** Pairs raw, noisy sequencing data directly with exact tissue- and cell-matched reference networks to eliminate validation mismatches.
- **Unbiased Model Benchmarking:** Establishes a standardized, uniform platform to objectively test, compare, and replicate different deep learning GRN inference algorithms.
- **Integrated Graph Analytics:** Features built-in structural analysis tools to compute network centrality metrics (e.g., PageRank, degree centrality) and map TF-pathway associations.

Some Ideas

- **Learnings from Past Models:** scMultiomeGRN and scTFBridge proved that integrating paired RNA and ATAC data is crucial for mapping physical regulation. However, both struggled with over-fitting and generalization due to noisy, mismatched, or computationally inferred validation baselines.
- **The Architecture (MEvD-GRN):** Proposing a deliberately compact, dual-modality Graph Neural Network. It utilizes a "Gated Fusion" module to dynamically weigh RNA expression levels against ATAC chromatin accessibility, directly addressing the multi-omic sparsity problem.
- **Leveraging SC-MO-GRN-DB:** Instead of just using the database for static benchmarking, we are using its tiered experimental data to drive Curriculum Learning. The model is trained sequentially from broad to strict biological evidence: Localization (ChIP-seq) -> Perturbation (CRISPR/KO) -> Dual-Evidence (Intersection).
- **Hypotheses:** By stepping the model from "easy/broad" binding data to "hard/strict" causal data, can we mathematically prevent the over-smoothing and generalization failures seen in previous GNN architectures? Can a lightweight model (~125K parameters) outcompete massive, computationally heavy baselines simply by utilizing a biologically smarter, evidence-aware training regimen?



THANK YOU!

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